Chapter 9

Achieving Operational Excellence and Customer Intimacy: Enterprise Applications

Video Cases

- Case 1: Maersk Develops a Global Shipping Management System
- Case 2: Life Time Fitness Gets in Shape with Salesforce CRM

https://www.youtube.com/watch?v=A7AEc-B2PKQ

Learning Objectives

9.1 Understand ERP Systems.

9.2 Understand SCM Systems.

9.3 Describe global SCM challenges.

9.4 Understand CRM Systems.

9.5 Discuss enterprise application challenges.

9.6 Describe how enterprise applications are using AI.

9.7 Understand how the information in this chapter can help your career.

TrueBlue Speeds Ahead with Enterprise Applications

- Problem
  -
  -
- Solutions

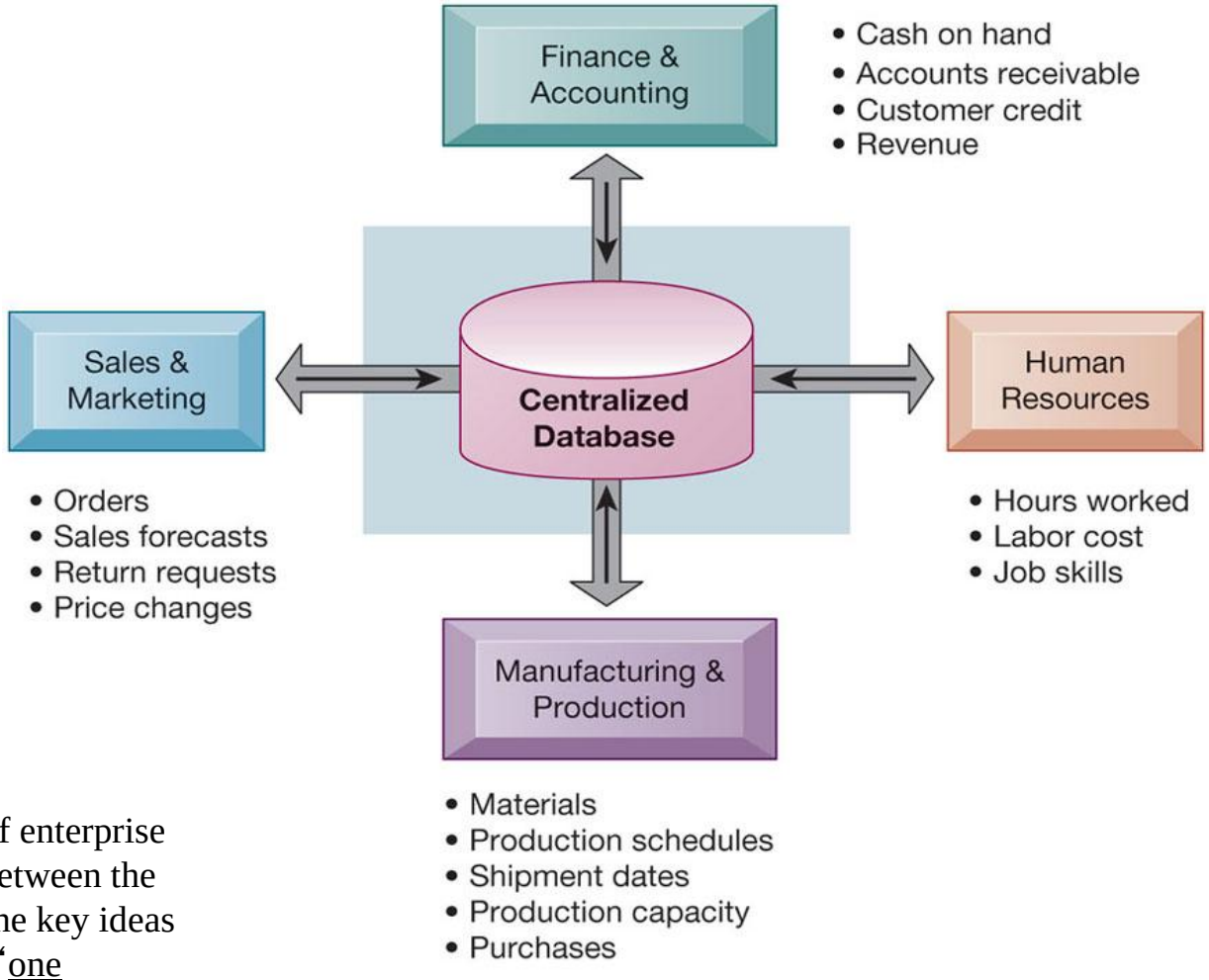
9.1 Understand ERP Systems

- Also known as Enterprise resource planning (ERP) systems
- Based on a suite of integrated software modules and a common central database
- Collects data from many divisions of firm for use in nearly all of firm's internal business activities
- Information entered in one process is immediately available for other processes **(see Fig. 9.1)**
- **AI and ERP Systems**: By integrating AI into their ERP systems, companies can improve organizational performance across various functions. AI-driven ERP systems can identify patterns in very large volumes of data.

=> Alcoa, which implemented enterprise software to eliminate redundant processes and systems, reduce requisition-pay-to-cycle time, reduce 20 percent of worldwide costs. This illustrates the tremendous value that an effective implementation of enterprise software can have.

Figure 9.1 How Enterprise Systems Work

Enterprise systems feature a set of integrated software modules and a central database that enables data to be shared by many different business processes and functional areas throughout the enterprise.



This graphic illustrates the function of enterprise software to integrate and share data between the different business functions. One of the key ideas of enterprise solutions is that there's "one company, one database" and not a collection of disconnected databases.

ERP Software

- **Built around thousands of predefined business processes that reflect best practices**
  - Finance and accounting
  - Human resources
  - Manufacturing and production
  - Sales and marketing

-> Table 9.2

- ERP systems are very complex, and extensive customization may degrade system performance, compromising the information and process integration that are the main benefits of the system.
- Companies must change the way they work to conform to the business processes defined by the system's s/w.

- **To implement, firms**:
  - Select functions of system they wish to use
  - Map business processes to software processes
    - Use software's configuration tables for customizing

This slide describes the functions in an enterprise software package and how it would be implemented by a firm. Think why it is typically best to perform only minimal changes to enterprise software, and instead, change the way the firm works in order to conform to the software's business processes.

Business Value of Enterprise Systems

- Increase operational efficiency
- Provide firm-wide information to support better decisions
- Enable rapid responses to customer requests for information or products
- Include analytical tools (to use the data the system captures) to evaluate overall organizational performance and improve decision-making
- Allow senior management to find out easily and at any moment how a particular organizational unit is performing

This slide discusses the business values of enterprise systems. What does it mean for the firm that enterprise systems enforce the use of common standardized definitions and formats for data by the entire organization?

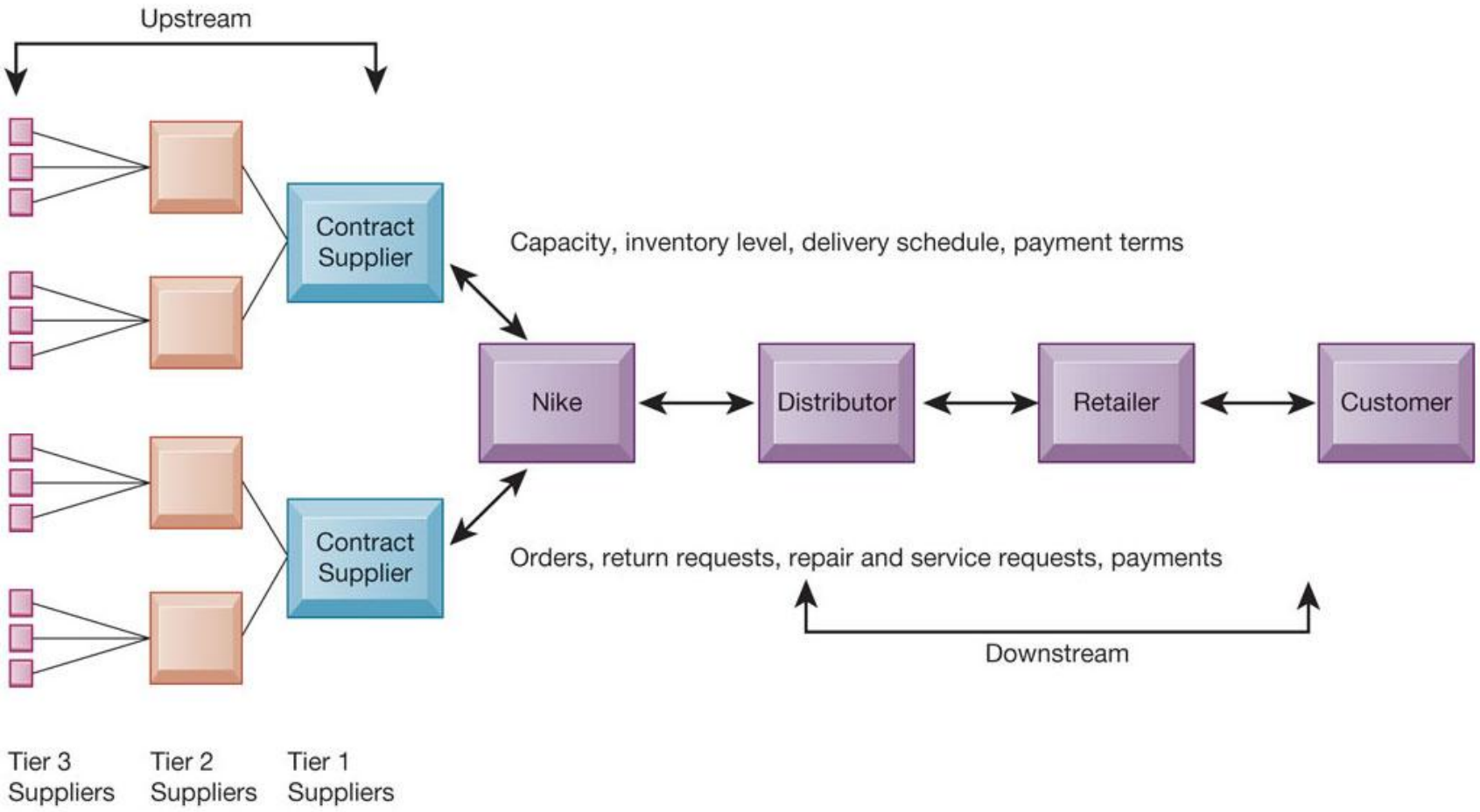
9.2 Understand SCM Systems

The Supply Chain

- Network of organizations and processes for:
  - Procuring materials
  - Transforming materials into products
  - Distributing the products
- Upstream supply chain: Firm's suppliers, suppliers' suppliers, processes for managing relationships with them
- Downstream supply chain: Organizations and processes responsible for delivering products to customers
- Internal supply chain processes (see p.325)

Supply chain management software is a type of enterprise software for managing complicated supply chains that may include thousands of suppliers. e.g. Nike's sneakers' supply chain. What types of firms are in the upstream supply chain? In the downstream supply chain?

Figure 9.2 Nike's Supply Chain



This figure illustrates the major entities in Nike's supply chain and the flow of information upstream and downstream to coordinate the activities involved in buying, making, and moving a product. Shown here is a simplified supply chain, with the upstream portion focusing only on the suppliers for sneakers and sneaker soles. From the contract suppliers a two-way path is depicted to illustrate the downstream process.

Supply Chain Management

- **Inefficiencies cut into a company's operating costs**
  (in the supply chain) cut into= interrupt
  - Can waste up to 25 percent of operating expenses
- **Just-in-time strategy**
  - Strategy that aims to have components exactly at the moment arrive as they are needed and finished goods shipped after leaving assembly line
- **Safety stock**: buffer for lack of flexibility in supply chain
- **Bullwhip effect**

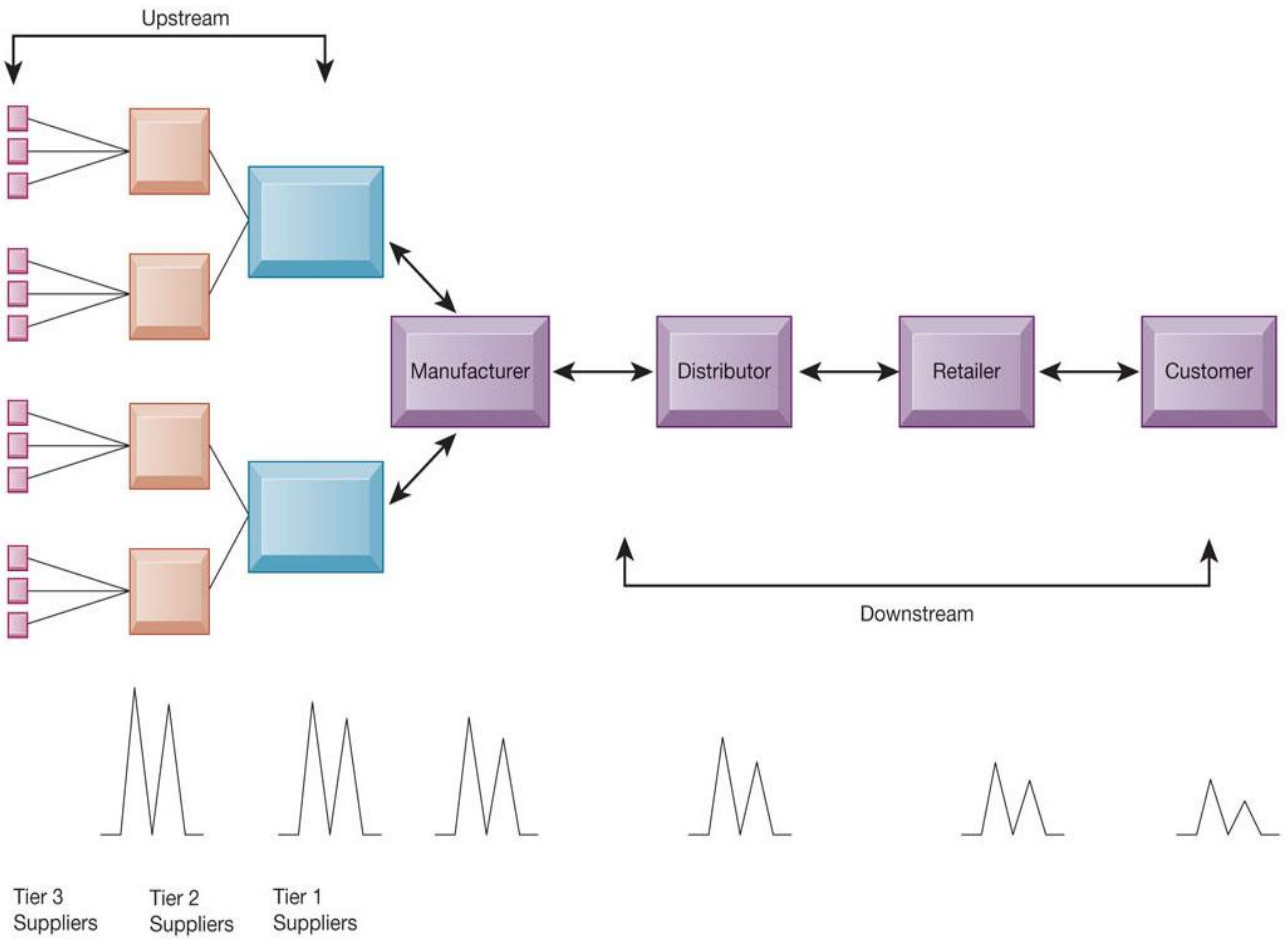
<-> low fill rates also costly because business may be lost form canceled orders

  - Information about product demand gets distorted as it passes from one entity to next across supply chain

This discusses the effects of timely and untimely information on a supply chain. Think what causes inefficiencies in a supply chain (parts shortages, underutilized plant capacity, excessive finished goods inventory, high transportation costs). These are caused by untimely or inaccurate information. Perfect information can result in a just-in-time strategy. Because of uncertainties, manufacturers keep a safety stock. Why is this an inefficient result? Another effect of uncertainties is the bullwhip effect. * Bullwhip effect is tamed by reducing uncertainties about demand and supply because all members of the supply chain have accurate and up-to-date information.

Figure 9.3 The Bullwhip Effect

Inaccurate information can cause minor fluctuations in demand for a product to be amplified as one moves further back in the supply chain. Minor fluctuations in retail sales for a product can create excess inventory for distributors, manufacturers, and suppliers.



This graphic illustrates how a slight rise in demand by customers for an item will cause successive participants to slightly overstock related inventory "just in case". These changes are amplified as one moves back further in the system.

만약을 위해서

Supply Chain Management Software

- Supply chain planning systems
  - Model existing supply chain
  - Enable demand planning
  - Optimize sourcing, manufacturing plans
  - Establish inventory levels
  - Identify transportation modes
- Supply chain execution systems
  - Manage flow of products through distribution centers and warehouses to ensure that products are delivered to the right locations in the most efficient manner

*** demand planning**

e.g., WMS

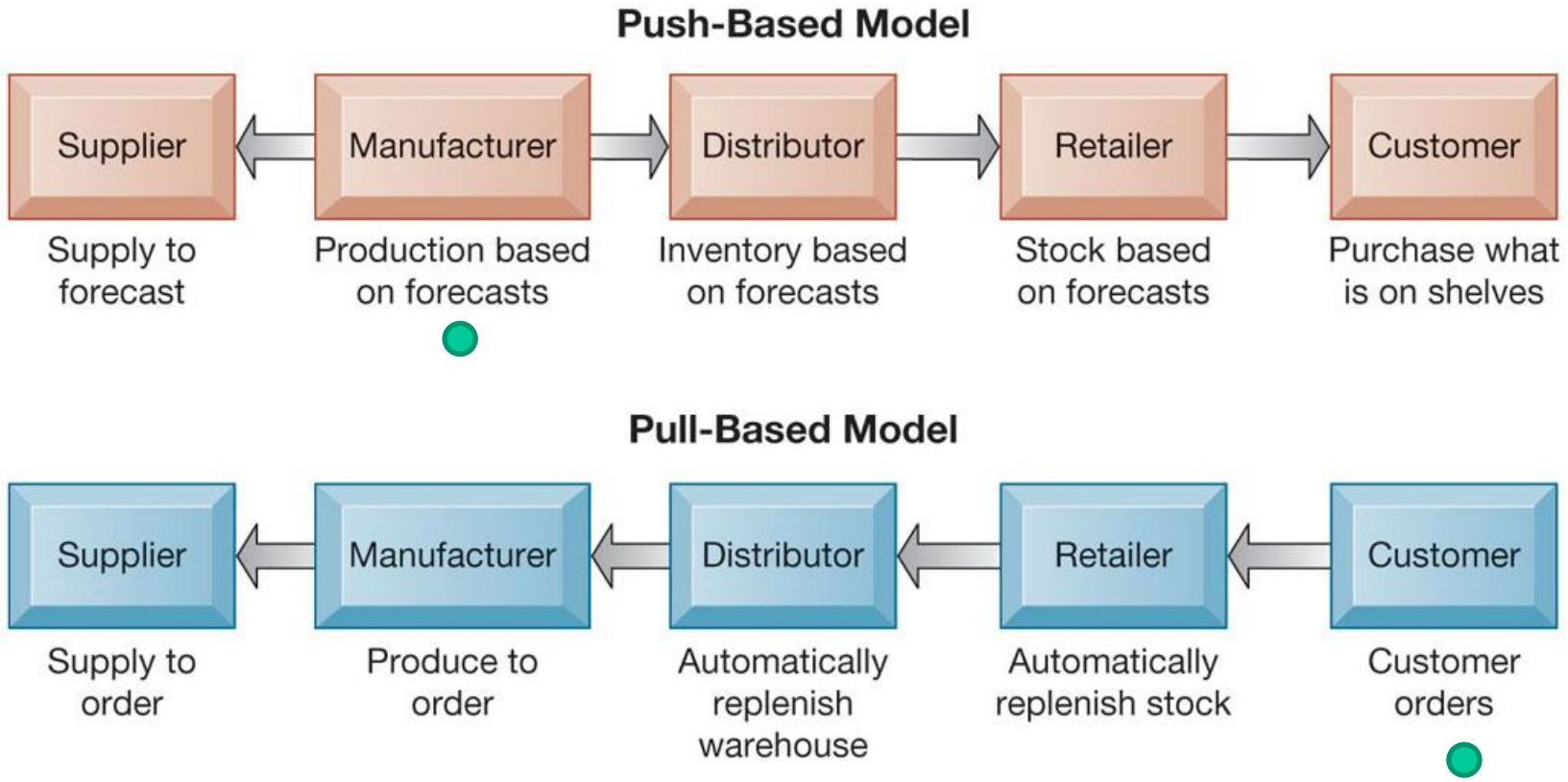
=> This slide discusses the two main types of supply chain management systems. Explain and describe the concept of "demand planning." It determines how much product a business needs to make to satisfy all its customers' demands.

Demand-Driven Supply Chains

* This slide discusses the fact that SCM systems facilitate efficient customer response, allowing the workings of the business to be driven more by customer demand, moving **from push-based, sequential supply chains to pull-based, concurrent supply chains** of production. Think for examples from the text of the pull-based model (Walmart, Dell).

- **Push-based model (build-to-stock)**
  - Earlier SCM systems
  - Supply chain driven by forecasts or best guesses of demand for products, rather than what customers order; products are "pushed" to customers
- **Pull-based model (demand-driven; build-to-order)**
  - Supply chain driven by actual customer orders or purchases; Web-based tools
  - Customer orders trigger events in supply chain
- **Internet enables move from *sequential* supply chains to concurrent supply chains**
  - Complex networks of suppliers can adjust immediately to changes in schedules or orders

Figure 9.4 Push- Versus Pull-Based Supply Chain Models



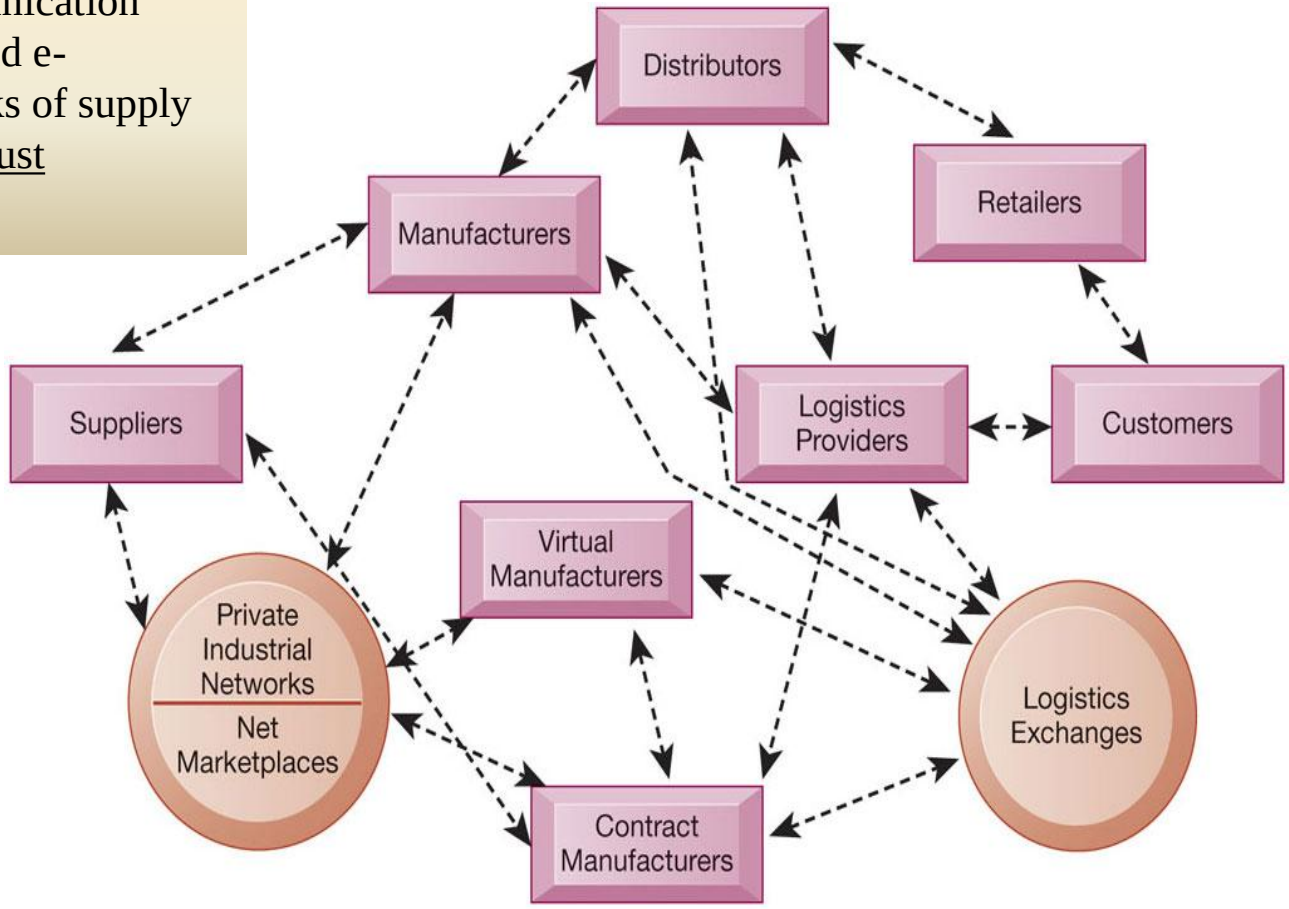
The difference between push- and pull-based models is summarized by the slogan "Make what we sell, not sell what we make."

* Think over if there are some industries uniquely suited to the pull-based model.
How about automobiles, appliances, and consumer durables? What are the drawbacks of a pull-based model?

Figure 9.5 The Emerging Internet-Driven Supply Chain

The emerging Internet-driven supply chain operates like a digital logistics nervous system. It provides multidirectional communication among firms, networks of firms, and e-marketplaces so that entire networks of supply chain partners can immediately adjust inventories, orders, and capacities.

This graphic illustrates the multidirectional communications within a future supply chain driven by the Internet. Private industrial networks and net marketplaces are discussed in Chapter 10. Private industrial networks are typically a large firm using an extranet to link to its suppliers and other key business partners. Net marketplaces are digital marketplaces based on Internet technology for many different buyers and sellers.



All the connections are depicted via bidirectional dashed arrows.

Business Value of Supply Chain Management Systems

- Match supply to demand
- Reduce inventory levels
- Improve delivery service
- Speed product time to market
- Use assets more effectively
- Reduced supply chain costs lead to increased profitability
  - Total supply chain costs can be 75 percent of operating budget
- Increase sales

* This slide examines the value of using SCM systems to businesses. Think how increased sales can result from a more efficient supply chain.

9.3 Describe Global Supply Chain Management Challenges

- Global supply chain issues <-> Before the Internet
  - Greater geographical distances, time differences
  - Participants from different countries
    - Different performance standards
    - Different legal requirements
- Internet helps manage global complexities
  - Warehouse management
  - Transportation management
  - Logistics
  - Outsourcing

* **AI and SCM**: AI is very useful for analyzing the proliferating amount of big data generated by modern global supply chains (p.331)

* This slide discusses the additional complexities experienced by global supply chains. Today's apparel industry relies heavily on outsourcing to contract manufacturers in China and other low-wage countries. Apparel companies are starting to use the Web to manage their global supply chain and production issues.

9.4 Understand CRM Systems

- Knowing the customer - The basis of competition is that who "owns" the customer and that customer relationships represent a firm's most valuable asset.
- In large businesses, too many customers and too many ways customers interact with firm
- **CRM systems**
  - Capture and integrate customer data from all over the organization
  - Consolidate and analyze customer data
  - Distribute customer information to various systems and customer touch points across enterprise
  - Provide single enterprise view of customers

* The use of CRM systems provides a single place to consolidate and analyze data about the customer. Think examples of the different types of customer data that would be captured by a firm. What types of valuable information does the firm need to know about its customers (most profitable, loyal customers). Anytime you call into a call center for help, you are dealing with some kind of CRM that tracks some or all of your interactions with a firm.

Figure 9.6 Customer Relationship Management (CRM)

CRM systems examine customers from a multifaceted perspective. These systems use a set of integrated applications to address all aspects of the customer relationship, including customer service, sales, and marketing.



This graphic illustrates the functions found in the integrated applications of a CRM system.

- A segmented circle explains the concept of customer relationship management.
- The circle is divided into three parts with an ellipse representing the customer at the center of the circle.
  -> Each of the parts belong to sales, marketing, and service respectively.

Customer Relationship Management Software (1 of 2)

- **Packages range from niche tools to large-scale (full-scale) enterprise applications**
- **More comprehensive packages have modules for**:
  - Partner relationship management (PRM) It uses many of CRM to enhance collaboration between a company and its selling partners.
    - Integrating lead generation, pricing, promotions, order configurations, and availability
    - Tools to assess partners' performances
  - Employee relationship management (ERM) It deals with employee issues that are closely related to CRM.
    - Setting objectives, employee performance management, performance-based compensation, employee training

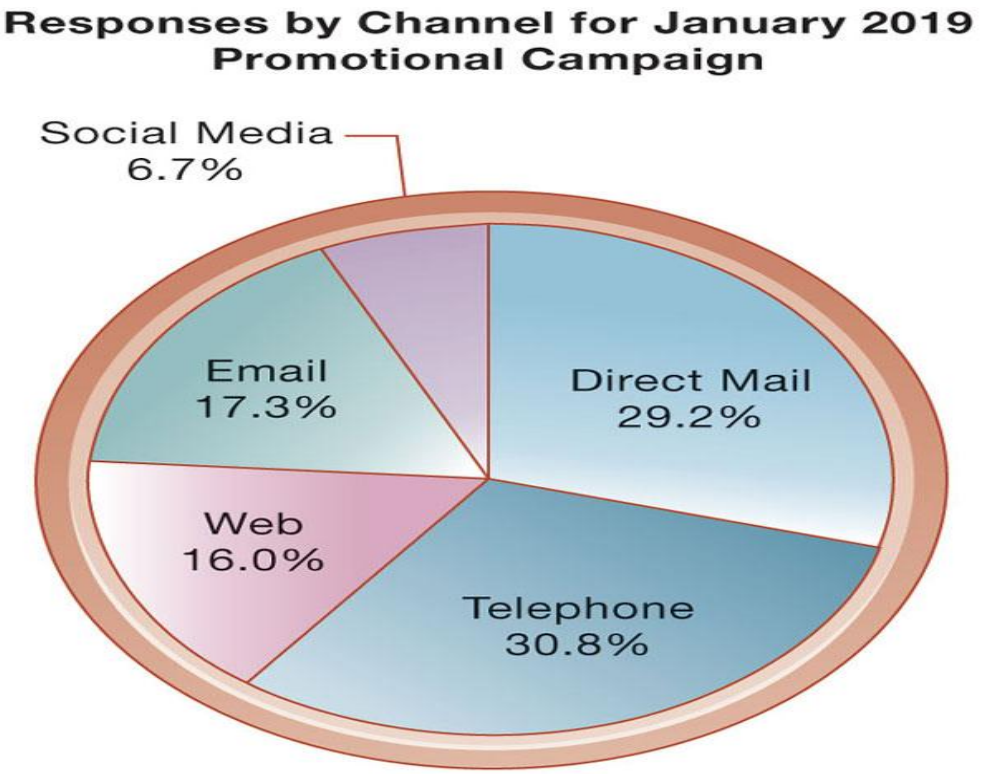
Customer Relationship Management Software (2 of 2)

- **CRM packages typically include tools for**:
  - **Sales force automation (SFA)** helps sales staff increase productivity by focusing sales efforts on the most profitable customers.
    - Sales prospect and contact information
    - Sales quote generation capabilities
  - **Customer service** provides information and tools to increase the efficiency of call centers, help desks, and customer support staff.
    - Assigning and managing customer service requests
    - Web-based self-service capabilities
  - **Marketing** supports direct-marketing campaigns by providing capabilities
    - Capturing prospect and customer data, scheduling and tracking direct-marketing mailings or e-mail
    - Cross-selling

      -> The marketing of complementary products to customers cf. up-selling

Figure 9.7 How CRM Systems Support Marketing

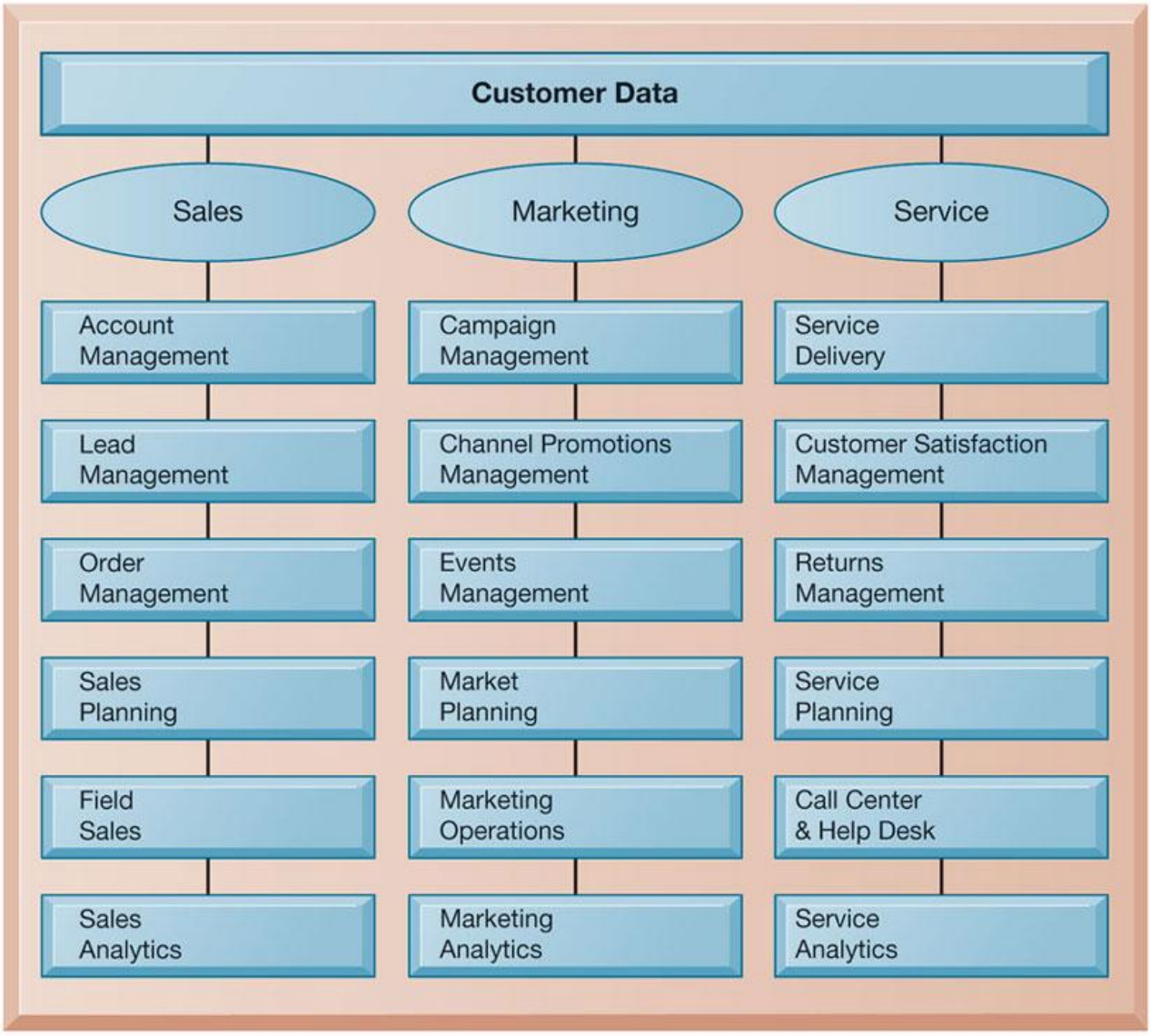
Customer relationship management software provides a single point for users to manage and evaluate marketing campaigns across multiple channels, including e-mail, direct mail, telephone, the Web, and wireless messages.



* This graphic gives an example of how CRM systems can support marketing by analyzing and evaluating marketing campaigns. A pie chart presents data on how customer relationship management systems support marketing.

Figure 9.8 CRM Software Capabilities

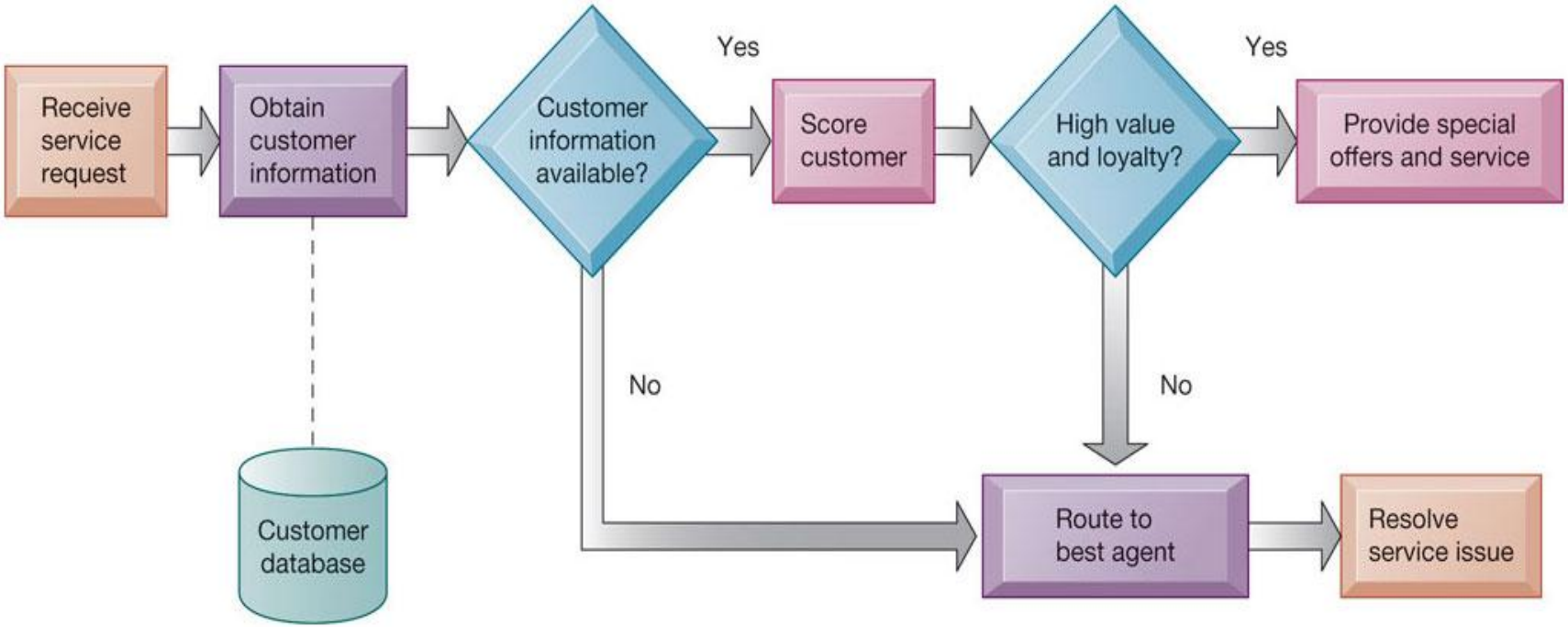
The major CRM software products support business processes in sales, service, and marketing, integrating customer information from many different sources. Included are support for both the operational and analytical aspects of CRM.



* This graphic illustrates the range of functions included in the sales, marketing, and service modules in a CRM package.

- CRM software is business-process driven, incorporating hundreds of business processes thought to represent best practices in each of these areas.
- To achieve maximum benefit, companies need to revise and model their business processes to conform to the best-practice business processes in the CRM software.

Figure 9.9 Customer Loyalty Management Process Map



This process map shows how a best practice for promoting customer loyalty through customer service would be modeled by customer relationship management software. The CRM software helps firms identify high-value customers for preferential treatment.

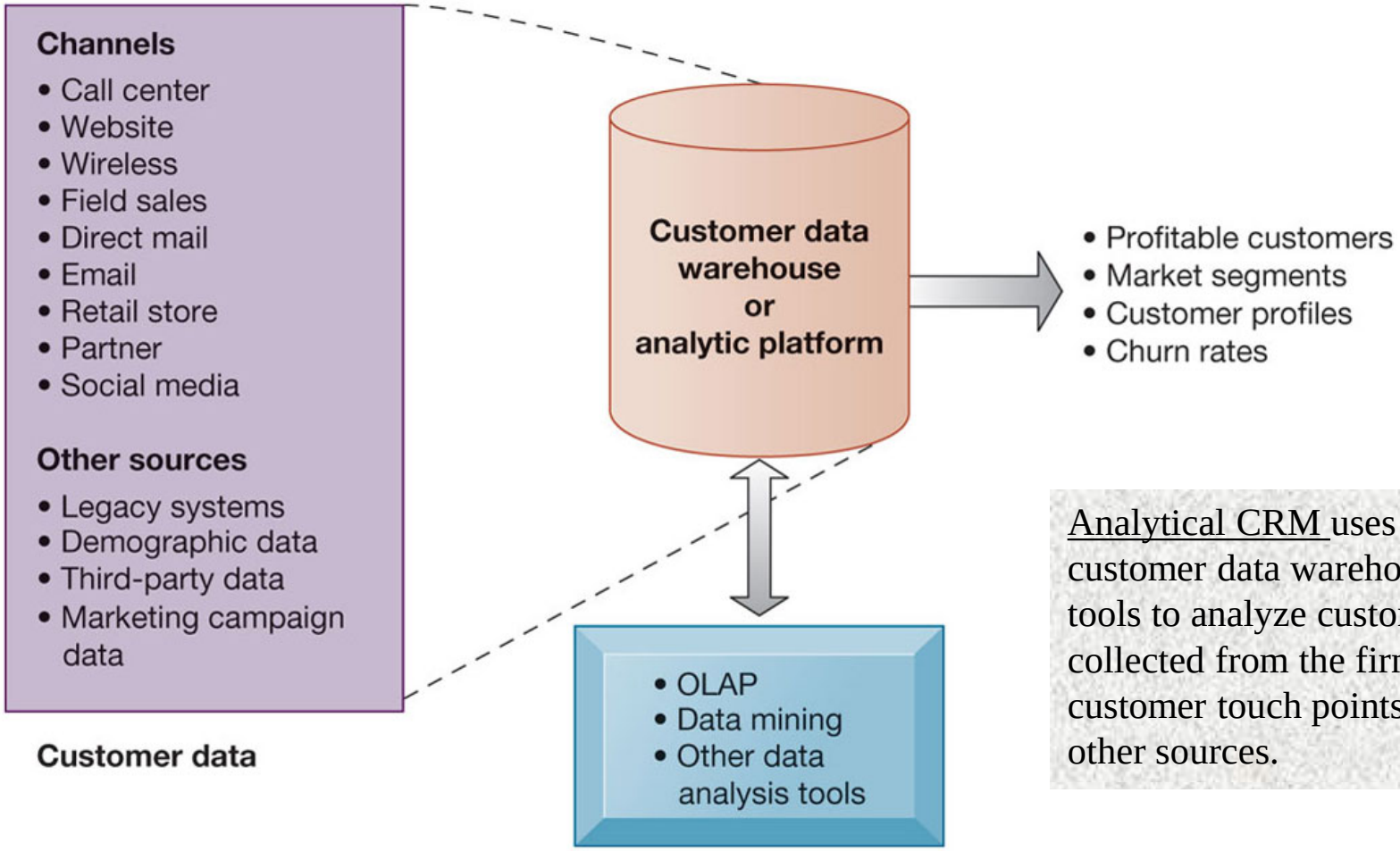
What kinds of information might indicate that a customer is loyal or high value?

Operational and Analytical CRM

- **Operational CRM:**
  - Includes **Customer-facing applications**
  - **Sales force automation, call center and customer service support**
  - **Marketing automation**
- **Analytical CRM:** includes applications that analyze customer data generated by operational CRM applications to provide information for improving business performance
  - **Based on data warehouses populated by operational CRM systems and customer touch points**
  - **Analyzes customer data (OLAP, data mining, etc.)**
    - **Customer lifetime value** (CLTV): another important output of analytical CRM

* Think how the **CLTV** is calculated (it is based on the relationship between the revenue produced by a specific customer, the expenses incurred in acquiring and servicing that customer, and the expected life of the relationship between the customer and the company.)

Figure 9.10 Analytical CRM Data Warehouse



Analytical CRM uses a customer data warehouse and tools to analyze customer data collected from the firm's customer touch points and from other sources.

* This graphic illustrates the main components of analytical CRM. Data is captured from various channels and fed into a customer data warehouse, where OLAP, data mining, and other analysis tools help identify profitable customers, churn rates, and so on.

Business Value of Customer Relationship Management Systems

- Business value of CRM systems
  - Increased customer satisfaction
  - Reduced direct-marketing costs
  - More effective marketing
  - Lower costs for customer acquisition/retention
  - Increased sales revenue
- **Churn rate**
  - Number of customers who stop using or purchasing products or services from a company
  - Indicator of growth or decline of firm's customer base

* It discusses the value to businesses of implementing a CRM system.
* How would marketing be made more effective by using a CRM system?

9.5 Discuss Enterprise Application Challenges

- Expensive to purchase and implement enterprise applications
  - Many projects experience cost overruns
  - Long development times
- Require Technology changes + Business process changes
- Organizational learning, changes
- Switching costs, dependence on software vendors

  - Integrating cloud applications
- Data standardization, management, cleansing

* Explain and apply with examples the concept of "switching costs."
  What problems could result from being dependent on a single software vendor?

Next-Generation Enterprise Applications (1 of 2)

Leading vendors include SAP, Oracle, Microsoft, and Netsuite.

- Enterprise solutions/ enterprise suites/ e-business suites:
  - Make applications more flexible, web-enabled, integrated with other systems
- Utilize Web services and SOA standards

SAP provides more than 500 Web services through its Web site.

- Include Open-source applications
- On-demand solutions
- Cloud-based versions

The undisputed global market leader in cloud-based CRM systems is Salesforce.com.

- Functionality for mobile platform
- Versions also available for small and medium-sized businesses

Next-Generation Enterprise Applications (2 of 2)

- **Social CRM**
  - Incorporating social networking technologies
  - Company social networks
  - Monitor social media activity; social media analytics
  - Manage social and web-based campaigns
- **Business intelligence (in Enterprise Applications)**
  - Inclusion of BI with enterprise applications
  - Flexible reporting, ad hoc analysis, “what-if” scenarios, digital dashboards, data visualization

9.6 Describe how enterprise applications are using AI

- EA vendors add BI (Business Intelligence) and AI tools to help managers obtain more meaningful information from the massive amounts of data these systems generate, including data from the IoTs.
- Table 9.5 provides examples of AI s/w tools from major EA vendors.

9.7 Understand how the information in this chapter can help your career